# Capstone Project Weekly Report

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| **Reporting Week:** | Week ending September 27, 2026 | **Sponsor:** | Deloitte |
| **Project Title:** | Air Conditioning Demand Forecasting & Supply Chain Optimization | **Report Owner:** | Vedika Sharma (PGID: 12510071) |
| **Team Members:** | Arvind, Hariharan, Vani, Vedika, Sai Prashanth | **Program:** | ISB AMPBA (Summer 2026) |

## Project Milestones

The team successfully engineered and validated the Enterprise Forecasting Engine, transitioning from baseline statistical modeling to an APICS-compliant, leakage-controlled machine learning tournament framework. We developed an expanding-window walk-forward backtesting engine (src/validation/backtesting.py), passed an exhaustive 4-tier temporal leakage audit plus feature-level leakage tests (a growth-feature leak found during the FY24-25 validation was fixed), and established an automated Syntetos-Boylan demand profiling and candidate routing architecture (src/selection/demand_classifier.py). Benchmarking was executed across 152 SKUs (70 ODU) and 5 regional branches across 4 distinct algorithmic families, and an auditable Champion/Challenger model registry (src/selection/champion_challenger.py) was instituted to govern production promotions.

## Progress Made in Current Week

### 1. Temporal Leakage Audit & Walk-Forward Validation Engine

- **Causal Walk-Forward Splits:** Engineered and validated src/validation/backtesting.py enforcing strictly chronological expanding-window rolling-origin splits where max(train) < min(test) across all validation folds, permanently eliminating shuffle/lookahead contamination.
- **Temporal Leakage Audit & Fix:** Temporal leakage controls (src/validation/leakage_audit.py, tests/test_leakage_audit.py): lag causality, rolling-window exclusion, split disjointness, plus two feature-level tests (no feature may depend on y(t); forecast-time features must equal training features). These tests exposed a leak in the 12-week growth feature, which was fixed.
- **Multi-Factor Stability Index:** Formulated an objective model stability score penalizing candidate models for cross-fold WAPE variance, worst-fold performance degradation, and directional bias inconsistency across rolling validation origins.

### 2. Syntetos-Boylan Demand Profiling & Dynamic Model Routing

- **Demand Intermittency Categorization:** Implemented the Syntetos-Boylan / Kostenko-Hyndman quadrant framework (src/selection/demand_classifier.py) using the standard cut-offs Average Demand Interval (ADI) = 1.32 and squared coefficient of variation ($CV^2$) = 0.49.
- **Portfolio Profiling Results:** Classified all 273 weekly ODU SKU × Branch series (training period to 1 Jul 2024): Erratic (62.3% of volume / 61 series), Lumpy (34.8% of volume / 166 series), Smooth (2.1% of volume / 9 series), Intermittent (0.8% of volume / 37 series).
- **Intelligent Candidate Routing:** Engineered heuristics routing sparse/lumpy profiles to Croston, SBA, and TSB; routing seasonal high-velocity series to SARIMA and ETS; and routing multi-feature series to LightGBM, XGBoost, and Random Forest.

### 3. Multi-Model Forecasting Tournament & APICS Zero-Safe Metrics

- **Unified Interface for 17 Candidate Models:** Standardized fit/predict interfaces across families: Baseline (Naive, Seasonal Naive, Moving Average, Exponential Smoothing), pooled Seasonal Index (region/branch month index × month-end factor; 8-week and 26-week level variants), Statistical (ARIMA; SARIMA = ARIMA on the seasonally adjusted series; ETS = damped-trend ETS(A,Ad,N) on the seasonally adjusted series), Intermittent (Croston, SBA, TSB), Machine Learning (Random Forest, Gradient Boosting, XGBoost, LightGBM) and an inverse-WAPE weighted Ensemble.
- **Recursive Autoregressive ML Prediction:** Implemented recursive forward walk-forward forecasting (predict_recursive) dynamically feeding simulated predictions back into autoregressive lag buffers to prevent feature matrix leakage.
- **Zero-Safe APICS Metrics Suite:** Built comprehensive error quantification (src/validation/metrics.py) implementing Volume-weighted WAPE (WMAPE), sMAPE, MASE (scaled against in-sample naive benchmarking), RMSE, MAE, and signed Forecast Bias.
- **Enforced Zero-Bounding Constraints:** Applied non-negativity post-processing across all statistical baseline outputs, eliminating negative demand artifacts (e.g., unbounded ARIMA predicting -104 units) identified in prior weeks.

### 4. Defensible Champion/Challenger Governance & Persistent Registry

- **Auditable Promotion Gates:** Implemented src/selection/champion_challenger.py requiring challengers to demonstrate >= 3.0% out-of-fold WAPE reduction, <= 15.0% absolute bias tolerance, and stability score >= 40.0 over incumbents to prevent model churn.
- **Persistent Model Registry:** Created atomic lifecycle transitions (ACTIVE_CHAMPION -> RETIRED with superseding lineage) persisted directly to JSON storage (src/monitoring/registry.py) with complete governance audit trails.

## Business & Model Observations

- **Measured Accuracy vs Baselines:** Champions are chosen per SKU × Branch by lowest mean RMSE over 4 rolling-origin folds of 12 weeks; no model family wins everywhere. On the FY24-25 holdout (2024-07-01 to 2025-06-30; static evaluation, models fitted once on data to 30 Jun 2024), engine accuracy (100% - WAPE) was 54.3% at South Region x Month versus 55.0% for seasonal naive; at SKU x Branch x Month it was -3.6% versus -55.9% for seasonal naive.
- **Intermittent Models for Lumpy Demand:** Lumpy series account for 34.8% of ODU volume (166 series) and Erratic series for 62.3%; Croston, SBA and TSB are added to the candidate set whenever more than 30% of weeks are zero and compete on the same RMSE ranking as every other model.
- **Champion/Challenger Prevents Churn:** Strict multi-fold stability scoring prevented volatile single-fold models from prematurely displacing proven champions, ensuring deployment defensibility.

## Updates / MoM from Sponsor & Faculty Mentor

### Deloitte Sponsor Updates:

- Deloitte confirmed alignment on the weekly SKU × Branch canonical modeling grain (8,951 rows).
- Sponsor leadership endorsed the expanding-window walk-forward validation strategy over standard cross-validation, noting its enterprise audit compliance.

### Faculty Mentor Updates:

- Faculty mentor reviewed the temporal leakage audit report and Syntetos-Boylan categorization breakdown.
- Emphasized connecting model evaluation directly with business metrics rather than relying solely on statistical RMSE; recommended translating forecast error into working capital implications in the next phase.

## Challenges

### Technical Challenges:

- Ensuring recursive multi-step forecasting in gradient boosted trees maintained strict lag integrity across a 12-to-24 week horizon without accumulating runaway error drift.
- Managing computational throughput during batch rolling-origin backtesting across 152 SKUs and 5 regional branches (resolved via vectorized matrix operations and caching).

### Non-Technical Challenges:

- Balancing model promotion criteria across conflicting stakeholder priorities: statistical purity vs. supply chain planner explainability.

## Key Highlighted Action Items & Next Week Tasks

### Key Highlighted Action Items:

- Translate model forecast errors into safety stock buffers and reorder point inventory policies.
- Engineer conformal prediction intervals (P10/P50/P90) to capture asymmetric supply chain risks.
- Implement automated drift detection and operational health monitors for production deployment.

### Tasks to Finish in Next Week:

- Develop the Probabilistic Inventory Optimization Engine (src/inventory/inventory_engine.py).
- Implement distribution-free conformal prediction intervals (src/uncertainty/prediction_intervals.py).
- Build What-If Scenario Stress Simulator (src/scenarios/simulator.py) for lead time shocks and promotional spikes.
- Implement statistical drift detection (PSI and KS test) and planner alert queues.